

Product Description

TCG-Gel-6000LOVE **Low Oil, Low Volatile Emissions** Thermal Gel is a thermally conductive single-part interface gel engineered to deliver effective heat dissipation across complex surface geometries and variable gap tolerances, while reducing stress transfer and maintaining reliable thermal contact in high-performance electronic systems. The silicone-based formulation combines high flow characteristics, low volatility, and electrically insulating properties to support stable long-term operation in reliability-critical applications.

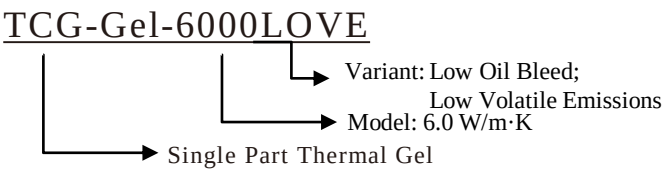
Key Features

- **Low Oil Bleed and Low Volatile Emissions**
- High conductivity and low thermal impedance
- Stable performance at elevated temperatures
- Demonstrated thermal cycling stability
- Electrically insulating to minimize component cross-talk
- Compatible with automated dispensing systems
- RoHS compliant

Applications

- ADAS Camera & LiDAR Modules
- Optical Tx/Rx & Photonics Systems
- 5G RRU & RF/mmWave Electronics
- SSD, DRAM & High-Density Memory Modules
- EV Battery & Automotive Electronics
- Sealed Sensor & Precision Electronic Assemblies

Product Naming



Application Guidelines

- Avoid exposure to open flame and direct sunlight.
- Ensure application surfaces are clean.
- Compatible with automated and manual dispensing systems.
- Avoid air entrapment during dispensing.

Storage and Shelf Life

- Store at 8°C to 28°C in a dry environment (RH<70%).
- Shelf life is 24 months from the date of manufacture.

Packaging Information

- Available in 30 cc (100 g) and 300 cc (1 kg) dual cartridge packaging
- Other sizes are available upon request

Property	Value	Test Method
Color	Gray	Visual
Flow Rate (g/min)	20	30cc tube, 0.090” nozzle, 90psi
Thermal Conductivity (W/m·K)	6.0	ASTM D5470
Minimum BLT (µm)	90	--
Density (g/cc)	3.5	ASTM D792
Breakdown Voltage (kV/mm)	>8.0	ASTM D149
Operating Temperature Range (°C)	- 50 to 200	--
Volatility (%)	0.13	ASTM E595
Flammability Rating	V-0	UL94
Volume Resistivity (Ω-cm)	>10 ¹³	ASTM D257
D3-D10 (ppm)	<50	Solvent Extraction
Oil Bleeding Size (mm)	< 1 mm	3 gram sample, 14 mm diameter, 100°C for 24 hours on grid paper

The data provided are typical values derived from laboratory testing and are intended for reference only. They should not be used as specifications or as a basis for design without independent validation.